

MODERN BIOPSYCHOSOCIAL PERSPECTIVES OF CRIMINAL BEHAVIOR

BioPsychoSocial

BioPsychoSocial

Biological: Genetics, Autonomic Nervous System, Hormones, and so on

Psychological: Psychopathy, Traits, Intelligence, and so on

Sociological: Economics, Environmental, Cultural Factors, and so on

**But really, this all
comes down to
measurement**



**What kind of
measurements?**

Family Studies

Family Studies

Twin Studies

Family Studies

Twin Studies

Adoption Studies

Family Studies

Twin Studies

Adoption Studies

Hormones

Family Studies

Twin Studies

Adoption Studies

Hormones

Neurotransmitters

Family Studies

Twin Studies

Adoption Studies

Hormones

Neurotransmitters

Brain Injuries

Family Studies

Twin Studies

Adoption Studies

Hormones

Neurotransmitters

Brain Injuries

Genetics

Family Studies

Twin Studies

Adoption Studies

Hormones

Neurotransmitters

Brain Injuries

Genetics

In other words, a lotttttt of stuff



Relax, I got you.

Heredity/Genetics vs. The Environment

In other words, sociologist based criminologist (*majority of the field*) vs. the biopsychosocial criminologist

Major argument was: how much of criminality is **inherited** from our parents (or ancestors) in comparison to how much of it is due to **only environmental factors**

From this point forward, *we will consider heredity and genetics as interchangeable, but will mostly use genetics*

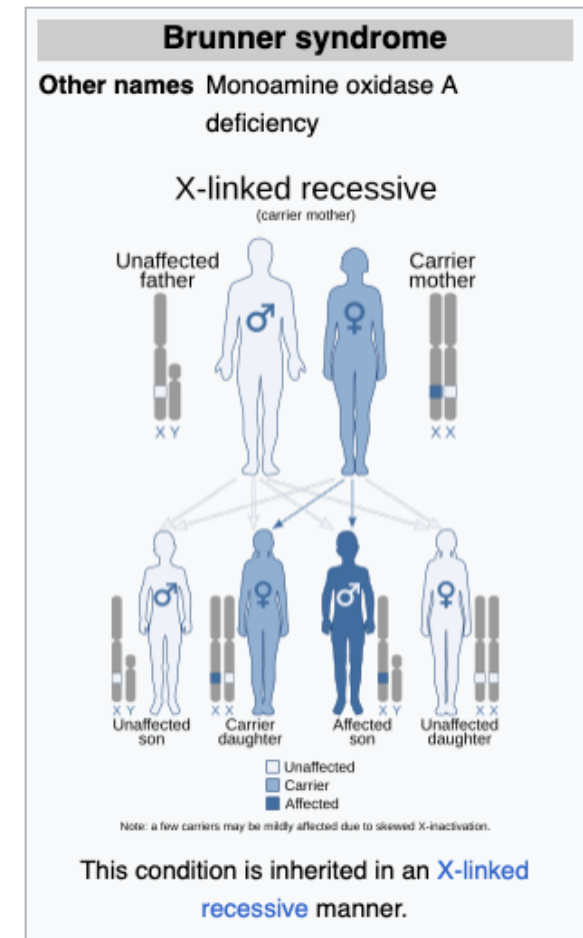
Family Studies

Brunner Syndrome (X-linked, MAOA gene mutation)

MAOA enzyme critical for breaking down neurotransmitters (serotonin, dopamine, norepinephrine).

Characterized by impulsive aggression (i.e., violent outbursts, antisocial behavior), mild intellectual disability, behavioral abnormalities (e.g., emotional dysregulation).

Landmark case linking genetics to behavior.



Adoption Studies

Compares adoptees to biological parents (genetic influence) and adoptive parents (environmental influence).

Genetic predisposition: Higher similarity to bio-parents (e.g., criminal behavior in adoptees correlates with bio-parents' history).

Environmental factors: Similarity to adoptive family (e.g., parenting style, trauma, socioeconomic status).

Adoptee with criminal behavior linked to bio-parents' history → **genetic risk**

Adoptee's aggression mirrors adoptive siblings exposed to abuse → **environmental risk**

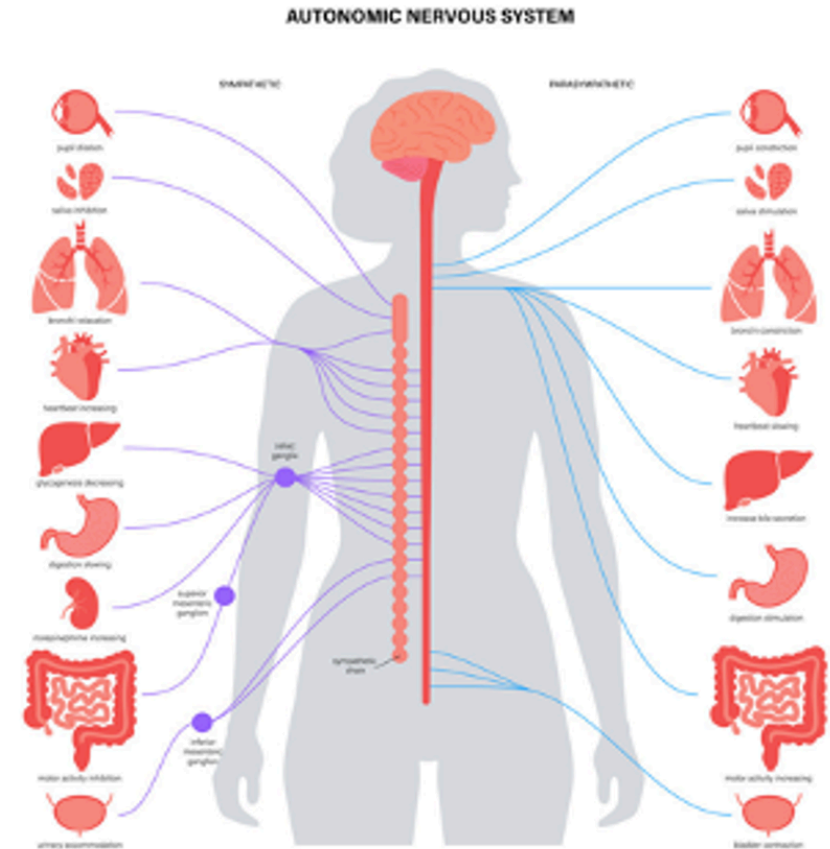
Physiology, Molecular Genetics, and Criminology

Autonomic Nervous System

A network of nerves regulating **unconscious bodily processes**

Helps the body adapt to environmental changes, particularly during **danger or stress**

Activates the “fight or flight” reaction to prepare the body for immediate action



ANS and Psychopathy

Low Resting Heart Rate Hypothesis:

Associated with increased antisocial behavior.

May reflect reduced fear, facilitating risk-taking and antisocial acts

Low Arousal Hypothesis:

Low ANS arousal may create an unpleasant 😞 physiological state.

Individuals may engage in stimulating or antisocial behaviors to increase arousal to optimal levels

Psychopathy





Psychopathy

A type of antisocial behavior characterized by constellation of traits:

Callousness: Lack of empathy or concern for others' feelings

Unemotionality: Absence of emotional depth or responsiveness

Manipulativeness: Skilled at exploiting others for personal gain

Impulsivity: Acting without forethought or consideration of consequences

Shallow Affect: Superficial or limited emotional expression

Resting Heart Rate

01:00

Check pulse

Watch the clock (15 seconds or 30 seconds) and count heart beats and multiply by 4 or 2

Write it down



Survey

01:00

The Inventory of Callous-Unemotional Traits

Higher scores  indicate higher Callous-Unemotional Traits!

The ICU has three subscales: Callousness, Uncaring, and Unemotional

Take your survey

Score and Analyze

What was your heart rate at rest?

What was your score?



**If you have a lower
heart rate at rest and
higher callousness
scores chances
are...**

Drum Roll



**You're just a hungry
college student who
wants this class to
end so you can go
have lunch 🤪**

RECAP

Hand back Exam!

News  incoming!!!

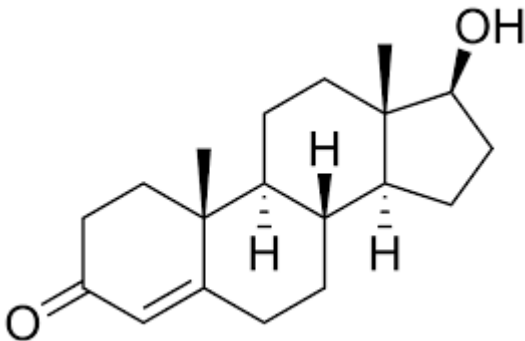
- I dropped twin studies from the lecture, won't be on any quiz or exam! Woohooooo!

Hormones

Hormones are the body's chemical messengers that travel through the blood stream (think long range 🚀)

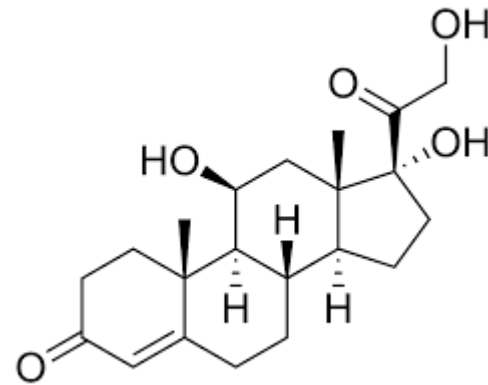
Testosterone

The major sex hormone in males, essential to the development of male growth and masculine characteristics



Cortisol

The body's main stress hormone, involved in the 🏃 or ✈️ response.



Hormones and Crime

Testosterone

Linked to aggression, dominance, and risk-taking behaviors, specifically violent and **impulsive** crimes (e.g., assault).

Cortisol

Low levels tied to reduced fear, poor impulse control, and **sensation-seeking**.

Interaction Effect

(**High** testosterone) x (**Low** cortisol) = aggression and rule-breaking (e.g., “dual hormone hypothesis”).

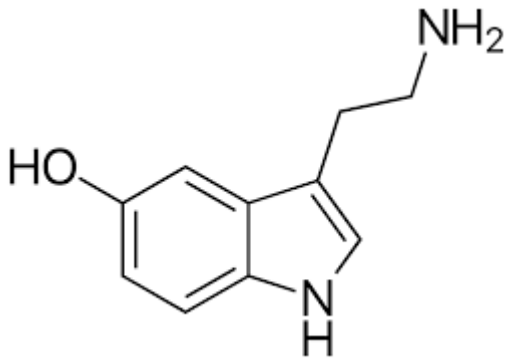
Here’s a cool paper on the role of **testosterone, cortisol, and criminal behavior in men and women**.

Neurotransmitters

Chemical messengers that transmit signals between neurons (think short range )

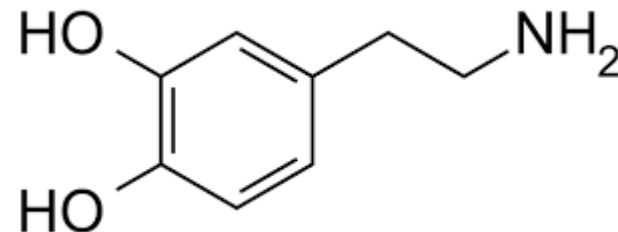
Serotonin

Often referred to as the 'feel-good' molecule - involved in mood regulation, sleep, appetite, learning, memory, and many other things.



Dopamine

Released during pleasurable activities, essential for motivation, attention, mood, and even physical movement (e.g., Parkinson's disease is a result degradation of the dopamine producing cells).



Neurotransmitters and Crime

Serotonin

Low serotonin levels have been associated depression & anxiety.

Mostly associated with crimes due to mental health incident which can be homicide and/or suicide. Also, drug-related offenses.

Very famous paper regarding serotonin aka 5HT

Dopamine

Hyperactivity is associated with addiction, compulsive behaviors, or crimes tied to [sensation-seeking](#).

Associated with antisocial personality disorders (e.g., psychopathy) and impulsive anti-social behavior.

Interaction Effect

([High](#) dopamine) x ([Low](#) serotonin) = heightened impulsivity and reward-driven aggression (e.g., “hot-tempered” vs. “cold-tempered” criminal acts)

Genetics x the Environment (GxE)

All the work showed that it is an **interaction** between the **molecular genetics** and the **environment**

It's never **just** genetics or **just** the environment. **Its both!**

And finally, as we'll see, genetics isn't always a **bad** thing.

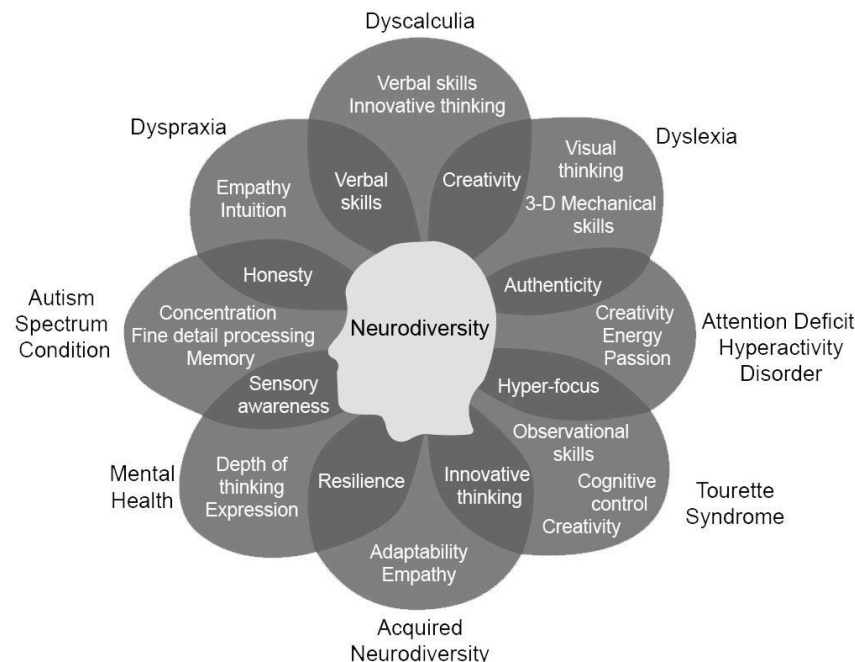
It really just make peoples more sensitive or more resistant to the environment they are.

Neurodiversity

The natural variations in brain function that shape how individuals process information, perceive the world, and interact with others. These differences—shaped by evolutionary and developmental influence.

People experience and interact with the world around them in many ways, with **no one “right” way of thinking**, learning, and behaving, and **differences are not deficits**.

Read more at [Harvard Health](#).



Differential Susceptibility

An evolutionary developmental theory proposing that genetic variation influences environmental sensitivity, shaping neurodiversity



Sensitivity and Neurodiversity

‘Dandelion’ Children

Children who are less reactive to their environments and are not as strongly affected by all experiences, both positive and negative

Can thrive in a wide range of circumstances and more resistant to environmental variation

‘Orchid’ Children

Children who are more sensitive to environmental factors and are more affected by all experiences, both positive and negative

Can thrive in optimal circumstance that are less resistant to environmental variation.

When given optimal circumstance they exceed in the top percentile level

When not optimal, they may be more susceptible to harsh effects of the environment they are in, and thus can lead to negative behavior outcomes

Group Discussion!!!!

Group Discussion

10:00

Studies suggest that both psychiatric interventions (e.g., Selective serotonin reuptake inhibitors) and mindfulness-based/holistic approaches can reduce symptoms related to antisocial behavior disorder. Given limited funding, should prison rehabilitation systems prioritize:

A.) Pharmacological interventions (e.g., SSRIs) with social support programs

B.) Holistic interventions (e.g., meditation) with social support programs

Pick one and make an argument for your choice. Elect one speaker to present your point.

Policy

Early Intervention Programs

Evidence-based parenting support: Fund programs to promote sensitive, proactive parenting strategies (e.g., positive discipline, emotional regulation training).

Nutritional Interventions for Crime Prevention

Integrate omega-3 fatty acids (e.g., fish oils) into at-risk youth diets, linked to improved impulse control and reduced aggression in clinical trials.

Have a Great Break!

If you're going on vacation or doing a stay-cation for Spring Break, make some "responsible" regrets, you only live once!



